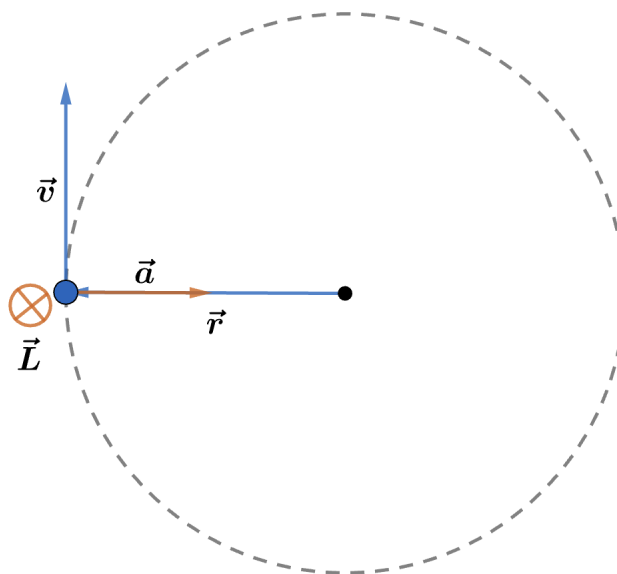


2014 F=ma Exam: Problem 1

Kevin S. Huang



In the driver's perspective, the position vector $\vec{r}$ points left from the center of the circle to the car. The velocity vector $\vec{v}$ always points forward. The angular momentum vector is given by

$$\vec{L} = \vec{r} \times \vec{p} \propto \vec{r} \times \vec{v}$$

By the right-hand rule, $\vec{r} \times \vec{v}$ points into the page (downwards for the driver) so that is also the direction of $\vec{L}$. Since the car is undergoing uniform circular motion, there is only centripetal acceleration directed toward the center of the circle. Thus, $\vec{a}$ points to the right relative to the driver. The answer is E.